

Total No. of Questions : 8]

SEAT No. :

PE-4283

[Total No. of Pages : 3

[6582]-55

S.E. (Information Technology Engg.)
PROCESSOR ARCHITECTURE
(2019 Pattern) (Semester - IV) (214451)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates :

- 1) Answer Q. No. 1 or Q. No. 2, Q. No. 3 or Q. N. 4, Q. No. 5 or Q. No. 6, Q. No. 7 or Q. No. 8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data, if necessary.

- Q1)** a) Write a short note on interrupt structure of PIC 18 microcontroller. [6]
b) Design interfacing of Relay and Buzzer to PIC 18F with the help of suitable diagram. Explain its working. [6]
c) Write the steps for Executing an interrupt in PIC 18F. [6]

OR

- Q2)** a) Draw an interfacing diagram for 16×2 LCD with PIC 18F microcontroller and explain its working. [8]
b) Explain any 3 sources of interrupt in PIC18F. [6]
c) Differentiate between interrupt and polling. [4]

- Q3)** a) List the steps involved in programming of PIC 18F with respect to PWM mode of CCP module. [6]
b) Draw and explain the stepper motor interfacing with PIC 18F. [6]
c) Write short note on I2C bus. [5]

OR

P.T.O.

Q4) a) Write the steps involved in programming compare mode of CCP1 module in PIC18F. [6]

b) How the speed of DC motor is controlled by PWM, explain in brief. [6]

c) Distinguish between synchronous and asynchronous serial communication. [5]

Q5) a) State the features of RTC. Explain function of following pins of DS 1306 [7]

i) SERMODE

ii) SDI

iii) SDO

b) Explain in detail the functions of ADCON1 SFR of PIC 18 microcontroller. [7]

c) List out the steps necessary for reading from EEPROM of PIC 18. [4]

OR

Q6) a) Explain the interfacing of LM34/LM35 with PIC18FXX for temperature measurement using on-chip ADC. [8]

b) Assuming that $R = 5\ \Omega$ and $I_{ref} = 2\ \text{mA}$ for DAC0808, calculate V_{out} for the following binary inputs: [6]

i) 10011001 (99H)

ii) 11001000 (C8H)

iii) 10001000 (88H)

c) Explain in detail the functions of following flags related to onboard ADC of PIC 18 microcontroller. [4]

i) GO/DONE

ii) ADON

- Q7)** a) Draw and explain ARM core dataflow model. [6]
b) Describe the ARM bus technology. [6]
c) Discuss the different exceptions in ARM processor. [5]

OR

- Q8)** a) What are the main features of ARM7 architecture? How it is different from pure RISC processor?. [6]
b) Write significance of special registers R13, R14 and R15 in ARM7. [6]
c) What are privileged and non-privileged modes? Write down the processor modes in ARM? [5]
